Transformers

Natural Language Processing

Daniel Ortiz Martínez

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Introduction

Transformers

- The transformer architecture (Vaswani et al. 2017) is based on the encoder-decoder framework
- Main idea: use attention mechanisms only
- Both the encoder and the decoder parts use stacked attention layers
- The transformer architecture is the foundation of current AI revolution

Large Language Models

- A language model (LM) defines a probability distribution over sequences of words
- LMs generate probabilities based on a set of parameters
- State-of-the-art LMs are based on the transformer architecture
- Large language models (LLMs) are LMs with >1B parameters

Transfer Learning

- Data scarcity may make difficult to apply machine learning (ML) to a certain task
- However, there may be a related task with plenty of data
- Transfer learning makes possible to transfer knowledge from the resource-rich task to the under-resourced one
- Transfer learning deemed as the next driver of ML commercial success

Transfer Learning

- Multiple approaches, but the main one used in LLMs is pre-training
- With model pre-training, a model trained for a general task is *fine-tuned* for a more specific or *downstream* task
- Pre-training features
  - Large and diverse data requirements
  - Computationally expensive
  - Useful in different applications
- Fine-tuning features
  - Smaller data requirements from a specific domain
  - Computationally cheaper
  - Useful in a specific application

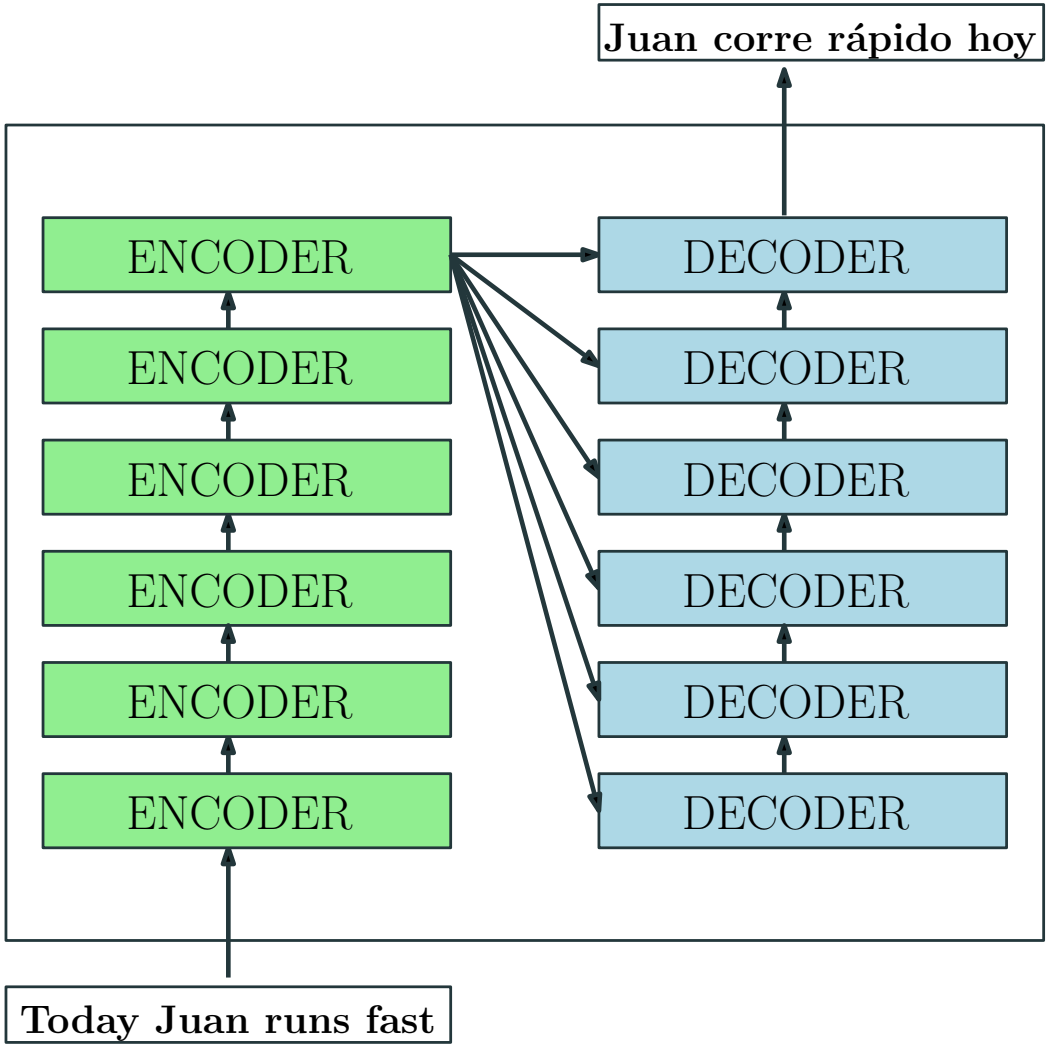
Zero-Shot and Few-Shot Learning

- Zero-shot and few-shot learning have gained very strong attention in connection with the development of LLMs
- **Zero-shot learning**: a classifier/learner is given samples from classes not seen during training and should predict the class they belong to
- **Few-shot learning**: a classifier/learner should predict the class of a set of samples for which only a few labeled training samples were available
- The ability of a model to perform zero-shot or few-shot learning is key for its widespread application due to supervised data scarcity

Transformer Architectures

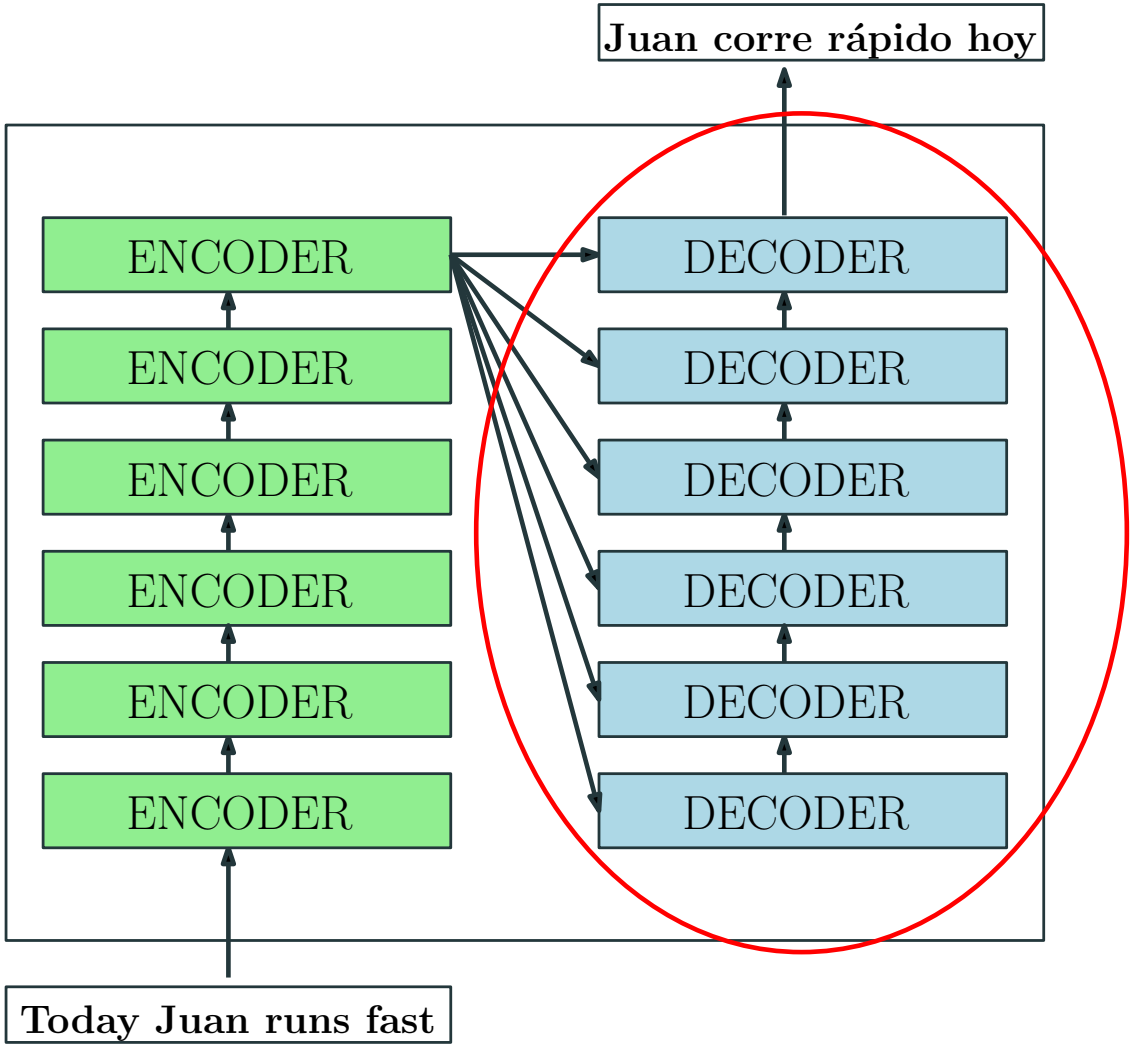
The Transformer Architecture (or Sequence-to-Sequence)

(Vaswani et al. 2017)



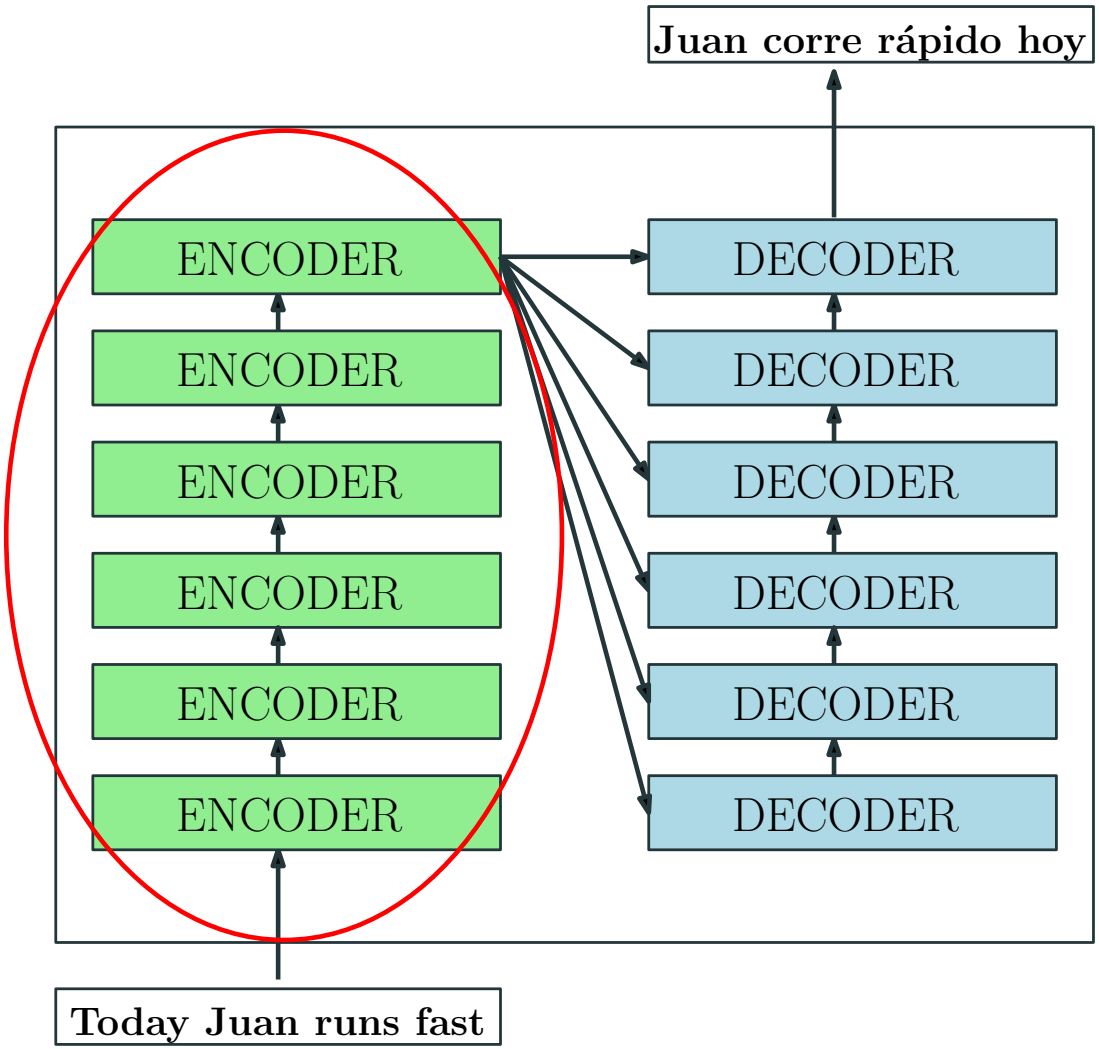
GPT

(Radford, Narasimhan, et al. 2018)



BERT

(Devlin et al. 2019)



Choosing the Right Transformer Architecture

- Choosing one architecture or another is tightly related to the ML task we need to solve
- BERT architecture: useful for classification tasks, less appropriate for text generation tasks
- GPT architecture: useful for text generation
- Whole transformer architecture: useful to solve sequence-to-sequence tasks (e.g. machine translation), can also be applied for classification

Transformer Implementations

Tokenization

- LLMs do not process raw text, but sequences of *tokens*
- A token can be a word, a subword unit, or a character
- The vocabulary is the set of all possible tokens; its size impacts model expressiveness and efficiency
- Popular tokenization algorithms:
  - **Byte-Pair Encoding** (BPE) (Sennrich et al. 2016): iteratively merges the most frequent pair of symbols; used in GPT-2, RoBERTa
  - **WordPiece**: similar to BPE but uses likelihood instead of frequency; used in BERT
  - **SentencePiece**: language-agnostic, operates on raw Unicode; used in T5, LLaMA
- Tokenization affects model performance

BERT

- BERT (Devlin et al. 2019) uses the encoder part of the transformer
- Tasks solved by the pre-trained model:
  - **Masked language modelling** (MLM): predicts masked words in a text
  - **Next sentence prediction** (NSP): given two sentences, predict whether they are consecutive or not
- Tasks solved by fine-tuned models:
  - Text classification
  - Sentiment analysis
  - Question answering
  - ...
- Not that useful for text generation

RoBERTa

- RoBERTa (Liu et al. 2019) is a BERT model with some tweaks:
  - Different tokenizer encoding: BPE (Sennrich et al. 2016)
  - Training with larger batches
  - Dynamic masking: masked words vary from one epoch to another
  - The NSP objective is removed from the pre-training scheme
- Tasks solved by the pre-trained model:
  - **Masked language modelling** (MLM): predicts masked words
- Tasks solved by fine-tuned models: same as BERT
- RoBERTa is also not appropriate for text generation

Distilled BERT-like Models

- A technique called *distillation* can be used to obtain smaller BERT-like models with almost the same performance of their larger counterparts
- Distillation trains a small *student model* to mimic a larger *teacher model*
- The canonical distilled model is DistilBERT (Sanh et al. 2019)
  - trained to predict the same probabilities as the original BERT model
  - predicts masked words but no next-sentence objective
  - preserves similarity between the hidden states of the student and teacher models

SBERT

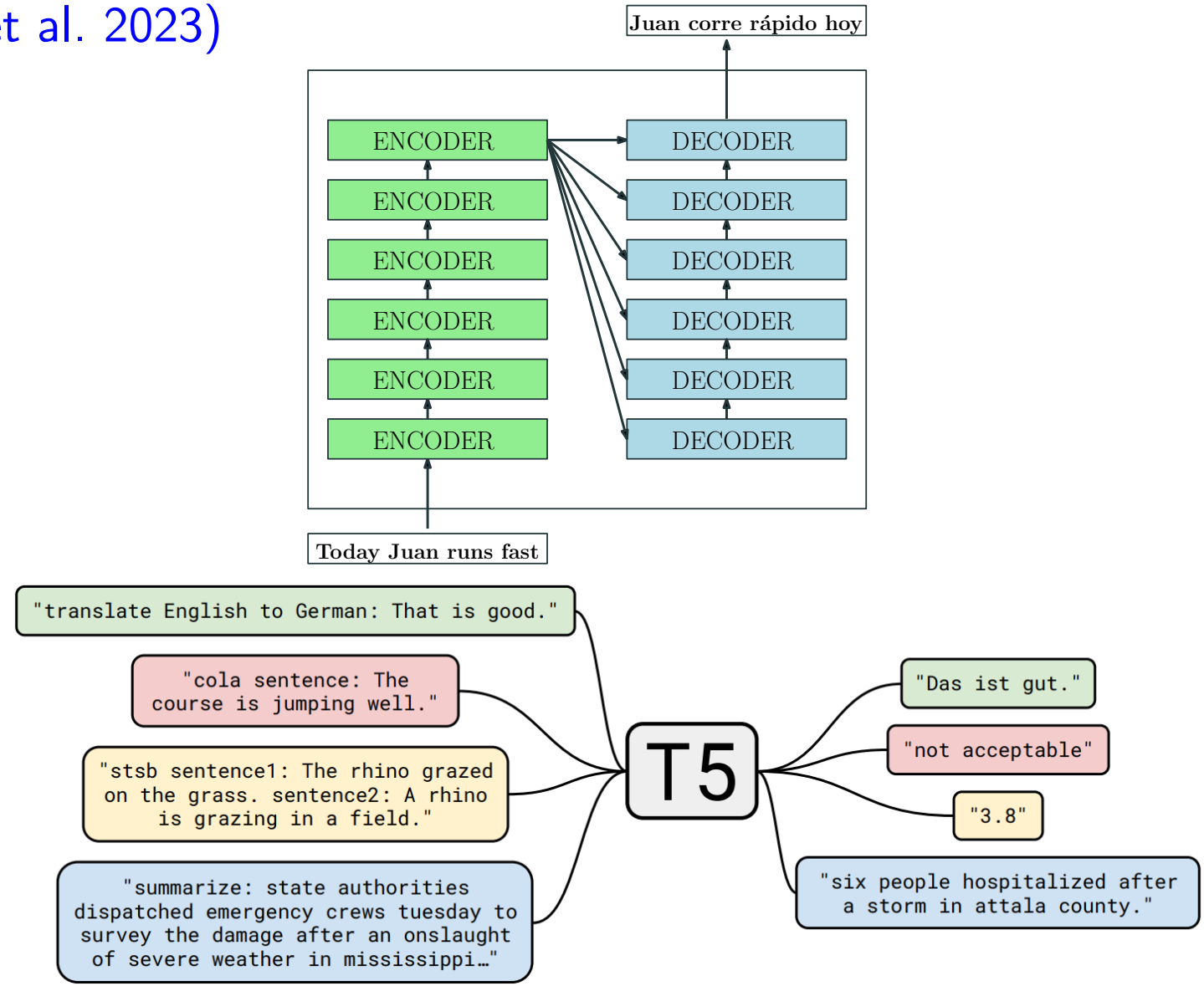
- Sentence-BERT (SBERT) (Reimers and Gurevych 2019) is a BERT model adapted for sentence embedding generation
- Conventional BERT internally generates information that can be used to obtain sentence embeddings
- However, those sentence embeddings are not of high quality
- SBERT solves this problem by means of two elements:
  - A pooling layer to the output of BERT
  - A fine-tuning process using supervised sentence pairs or triplets whose goal is to assign similar embeddings to similar sentences

BART

- BART (M. Lewis et al. 2019) follows a sequence-to-sequence architecture
- BART introduces a few network modifications with respect to the standard transformer architecture
- BART's pretrained model solves the task of reconstructing a previously corrupted or noised input
- Tasks solved by fine-tuned models: BERT-related tasks but also enables text generation tasks

T5 (Text-to-Text Transfer Transformer)

(Raffel et al. 2023)



Transformers for Text Generation

- Text generation is also called causal language modelling
- Models typically follow a GPT or a sequence-to-sequence architecture
- Two main types:
  - **Language model**: given a context predicts the next word
  - **Instruction model**: has been trained to follow instructions (instruction fine-tuning)
- Some examples:
  - OpenAI GPT-2 (Radford, Wu, et al. 2018)
  - LLaMA (Touvron et al. 2023)
  - Falcon[a]
  - Mistral[b]

[a] https://falconllm.tii.ae/

[b] https://mistral.ai/news/announcing-mistral-7b/

Transformers for Text Generation: Commercial Models

- GPT (OpenAI): dominant general-purpose model family; GPT-5 is state of the art in 2026
- Gemini (Google): strong multimodal capabilities; Gemini 2.5 Pro excels at interactive applications
- Claude (Anthropic): focus on AI safety; strong at long-document reasoning and analysis
- Mistral (Mistral AI): frontier-class performance at reduced cost; suitable for high-volume applications
- Grok (xAI): real-time data integration and current information retrieval
- DeepSeek (DeepSeek AI): competitive open-weight model developed at significantly lower cost; strong in technical reasoning

Transformers for Code

- Code can be seen as text belonging to a (non-natural) language
- There are many interesting tasks and applications related to code (code generation, code completion, code review, etc.)
- Wide range of transformer models working with code:
  - Code2Vec (Alon et al. 2018): code embedding generation
  - CodeT5 (Wang et al. 2021): T5 model, text to code, code completion and summarization
  - StarCoder (Li et al. 2023): LLM for code generation
  - WizardCoder (Luo et al. 2023): Code LLM with instruction fine-tuning
  - Also possible to apply general transformers: GPT, Gemini, LLaMA, etc.

Transformers for Biology: DNABERT

- Transformers can be useful in all those cases where we have sequential data that is categorical
- This type of data is common in biology (e.g. nucleotide or amino acid sequences)
- DNABERT (Ji et al. 2021) is a BERT-like model trained on a DNA genome reference
- Examples of downstream tasks:
  - prediction of promoters (locations where DNA transcription starts)
  - prediction of splice sites (locations involved in the maturation of RNA)
  - prediction of transcriptor factor binding sites (locations involved in the regulation of gene expression)

Transformers for Other Modalities

- Transformers cannot only be applied to textual modalities
- Transformers for images
  - Example: Visual Transformer (Dosovitskiy et al. 2021)
  - Survey: (Khan et al. 2022)
- Transformers for audio
  - Example: SpeechT5 (Ao et al. 2022)
  - Survey: (Latif et al. 2023)
- Transformers for video
  - Example: TimesSformer (Bertasius et al. 2021)
  - Survey: (Selva et al. 2023)
- Transformers can also combine modalities (multimodal)
  - Survey: (Xu et al. 2023)

Miscellaneous Topics

LLM Operation

- Operating LLMs is computationally expensive and frequently carried out with cloud computing resources
- However, there are initiatives to make LLM operation affordable in consumer hardware, even using CPUs
- A really nice collective effort is `llama.cpp`[c]
  - Supported platforms: Linux, Mac OS, Windows
  - Supported hardware: GPUs, CPUs, Hybrid GPU+CPU
  - Supported models: LLaMA, Mistral, Falcon, Starcoder, etc.
  - One interesting trick applied: quantization
  - A rich software ecosystem is being built around the package

[c] https://github.com/ggerganov/llama.cpp

LLM Training

- LLM training even more computationally expensive than LLM operation
- When fine-tuning a model, not all the weights may need to be substancially modified
- Low-Rank Adaptation (Hu et al. 2021) (LoRA) indirectly trains the dense layers of a given network by optimizing rank decomposition matrices of the dense layers' change during adaptation
  - the number of trainable parameters can be reduced by 10 000 times
  - the GPU memory requirements can be reduced by 3 times
- Quantized LoRA (Dettmers et al. 2023) (QLoRA) enables LoRA for quantized LLMs

LLM Evaluation

- LLM evaluation remains an open problem
  - What LLMs can do well?
  - Where LLMs fail?
- Two types of evaluation measures:
  - Automatic: objective, cheap, fast, typically quantitative
  - Manual: subjective, expensive, slow, quantitative or qualitative
- Benchmarking has become a popular approach, e.g. Open LLM Leaderboard

LLM Evaluation

- Some popular metrics:
  - ARC: grade-school level, multiple-choice science questions
  - HellaSwag: evaluates a model's common sense reasoning.
  - MMLU: covers a variety of academic topics across a variety of levels
  - TruthfulQA: model's ability to provide truthful and factual responses
- There are evaluation frameworks for LLMs, e.g. DeepEval
- Two survey papers on LLM evaluation: (Chang et al. 2023; Guo et al. 2023)

LLM Alignment: RLHF

- A pre-trained LLM predicts text but is not necessarily helpful, harmless, or honest
- **Reinforcement Learning from Human Feedback** (Ouyang et al. 2022) (RLHF) is the dominant technique for aligning LLMs with human preferences
- Three stages:
  - **Supervised fine-tuning** (SFT): fine-tune on human-written demonstrations
  - **Reward model training**: train a model to score outputs based on human preference comparisons
  - **RL fine-tuning**: use the reward model to further optimize the LLM via Proximal Policy Optimization (PPO)
- Used to train most modern instruction-following models

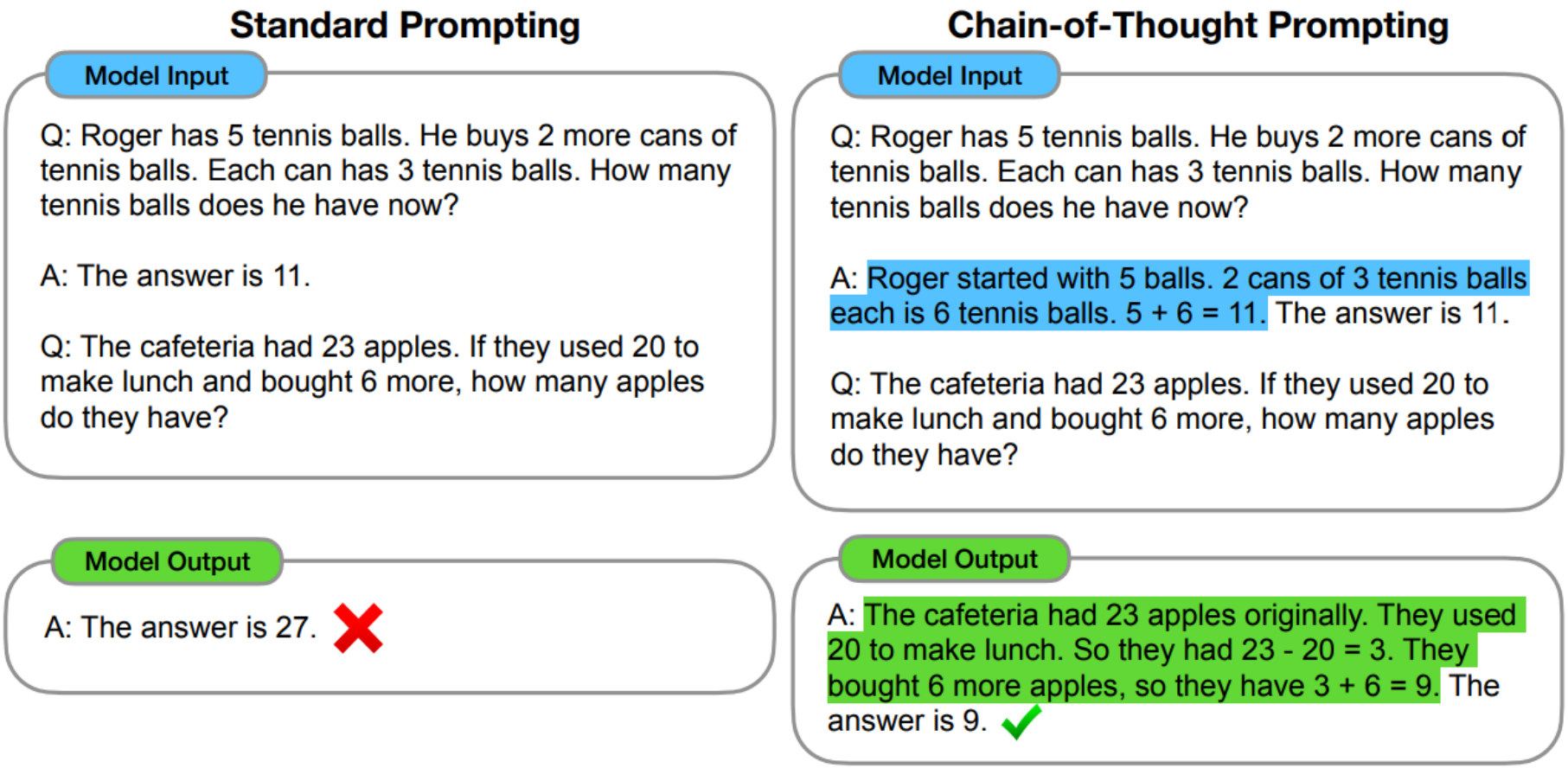
LLM Hallucinations

- LLMs can generate fluent and confident text that is factually incorrect: this is known as *hallucination*
- Two main types:
  - **Intrinsic**: contradicts the provided source or context
  - **Extrinsic**: introduces information not grounded in any source, which may be true or false
- Root causes:
  - Training data noise or gaps
  - Models optimized for fluency, not factuality
  - Knowledge cutoff: models ignore events after training
- Mitigation strategies: RAG, fine-tuning on factual data, TruthfulQA-style evaluation (Lin et al. 2022)

LLM Prompting

- Prompting may have a deep impact in the quality of LLM answers
- In-Context-Learning (Brown et al. 2020) (ICL) provides input-output pairs aiming that the LLM learns to perform a particular task by analogy
- A variant of ICL is called chain of thought (Wei et al. 2023) (CoT): CoT extends ICL by providing not only input-output pairs but also intermediate reasoning steps leading to the originally sought answers

LLM Prompting: CoT Example



NOTE: Figure taken from (Wei et al. 2023)

Retrieval-Augmented Generation (RAG)

- LLMs have a fixed knowledge cutoff and may hallucinate when asked about facts not seen during training
- **Retrieval-Augmented Generation** (P. Lewis et al. 2020) (RAG) addresses this by combining an LLM with an external knowledge source
- RAG pipeline:
  - **Indexing**: documents are split into chunks and stored as embeddings in a vector database
  - **Retrieval**: given a query, the most relevant chunks are retrieved by similarity search
  - **Generation**: the retrieved chunks are passed as context to the LLM, which generates a grounded answer
- Key advantage: knowledge can be updated without model retraining

Model Explainability

- Model explainability is a topic of growing interest in the field of machine learning
- Neural networks are black boxes
- When a prediction is made, it would be useful to know which sample features were relevant in the model decision
  - Better explainability of the model can make the user trust it more
  - Explainability information can also be used to troubleshoot the model predictions (real example: an image recognition system learned to classify images as horses by looking at a copyright tag that was exclusive to horse pictures, the problem was diagnosed by applying explainability techniques)

Model Explainability

- Two popular explainability techniques:
  - Shapley values (Shapley 1953)
  - Integrated gradients (Sundararajan et al. 2017)

- Main idea: calculate the contribution (or importance) of each feature to the network output for a particular sample

Model Explainability

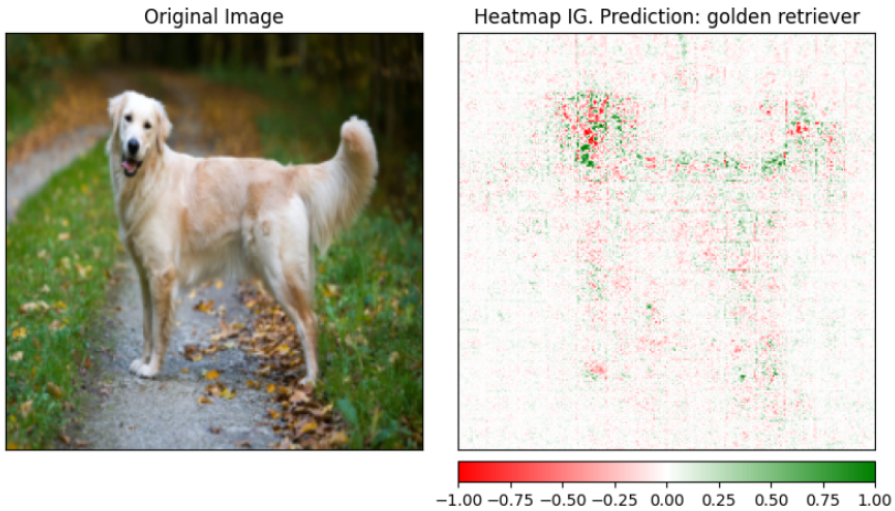
- Shapley values are more general and can be applied to any model, but they are computationally costly

- Integrated gradients can be applied to differentiable models only (any model trained with gradient descent, one example of non-differentiable model would be random forests), but they are faster to compute

- Implementations:
  - Shapley values: SHAP
  - Integrated gradients: Captum, Transformer Interpret

Model Explainability: Integrated Gradients Examples

- Example with images:



- Example with text:

Legend: Negative Neutral Positive

True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
POSITIVE	POSITIVE (1.00)	POSITIVE	2.02	[CLS] i love you , i like you [SEP]

LLM Agents

- An *LLM agent* is a system that uses an LLM as its core reasoning engine and can autonomously take actions to achieve a goal
- Key components:
  - **Planning**: decompose a complex task into sub-tasks (e.g., using CoT)
  - **Memory**: short-term (context window) and long-term (external storage or retrieval)
  - **Tools**: the agent can call external tools such as search engines, code interpreters, APIs, or databases
- Agents can operate in a loop: *observe* $\rightarrow$ *think* $\rightarrow$ *act* $\rightarrow$ *observe* ...
- Examples: OpenAI Assistants, LangChain agents, AutoGPT

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